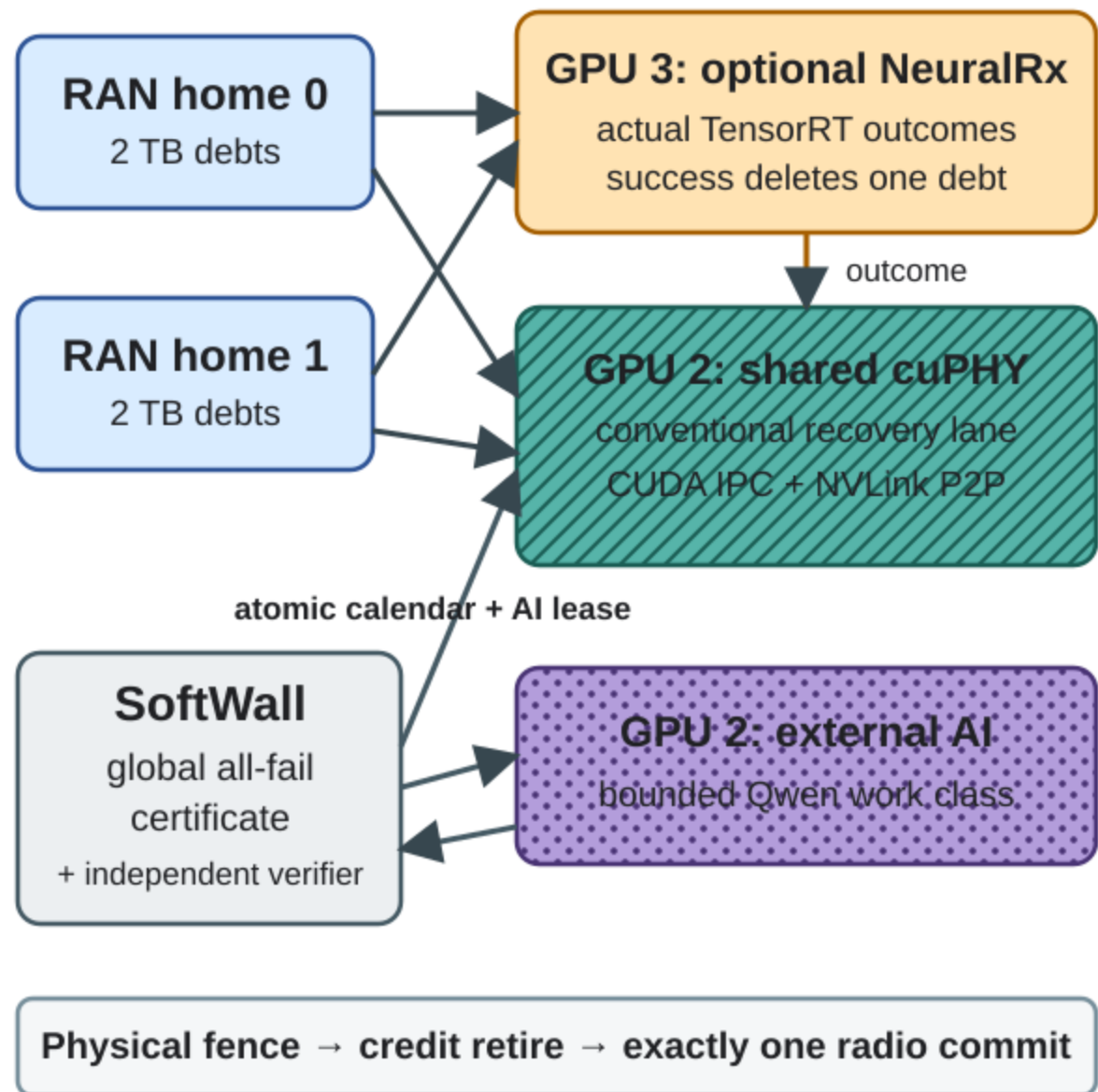
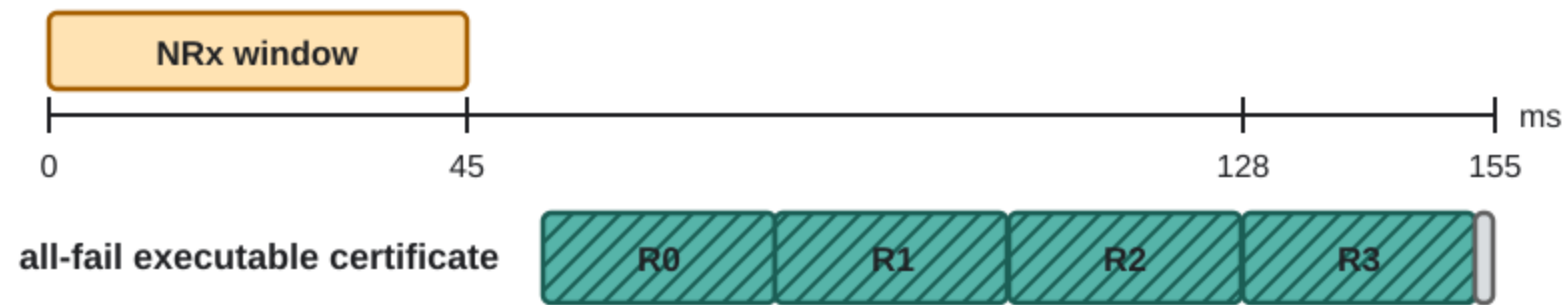


### (a) Certificate-preserving AI-RAN path

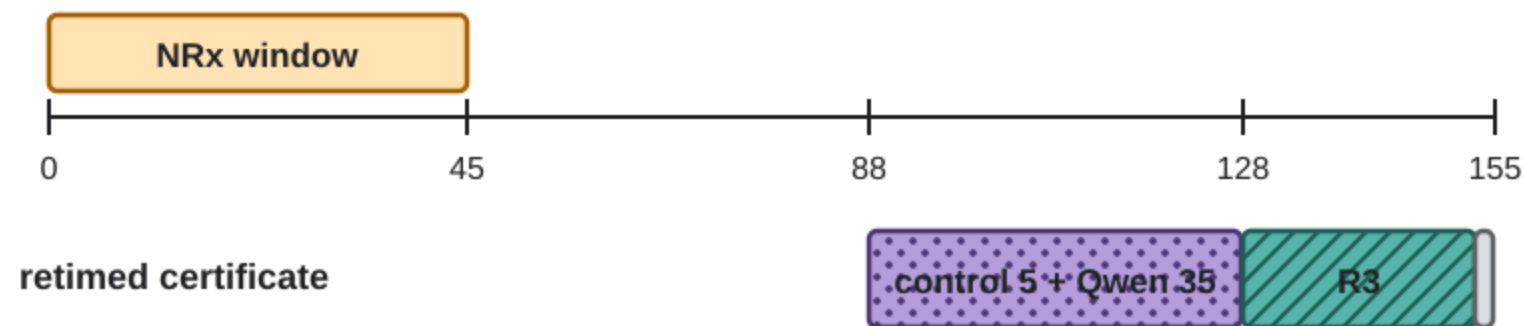


### (b) One certificate covers all failure subsets

Initial all-fail state: four unresolved debts



After three observed successes: delete three credits atomically



**decision at 88 ms**

$$40 + 25 + 2 = 67 \text{ ms} \rightarrow \text{QSU}$$

**decision at  $\geq 89$  ms**

$$40 + 25 + 2 > 66 \text{ ms} \rightarrow \text{QSN}$$

**QSU/QSN boundary is predicted before physical execution**